Evaluating Digital Twin Fidelity for Vacuum-Based Robotic Paper Manipulation in NVIDIA Isaac Sim

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***Abstract*—Paper manipulation is a challenging robotic task due to the highly deformable nature of paper and the complex interaction between suction grippers, material properties, and contact dynamics. Although physics-based simulators have become valuable tools for robotic system development, their ability to accurately reproduce paper manipulation has not been systematically evaluated. This work presents a real-to-simulation study of vacuum-based paper manipulation using a physical robotic platform and its corresponding digital twin implemented in NVIDIA Isaac Sim. The simulation model was calibrated through a systematic evaluation of deformable-body parameters, including material representation, stiffness, mesh resolution, grasp implementation, and solver iterations. Grasping experiments were then performed at nine representative grasp locations on an A4 paper sheet in both environments , using a single trial per position. Agreement between the real and simulated deformations was evaluated using geometric metrics, including planar dimensions, sagitta, and maximum height, as well as qualitative comparisons of deformation profiles. The study also identifies important limitations of the current simulation framework, particularly regarding vacuum interaction with porous paper surfaces and contact with the supporting surface. The findings indicate that Isaac Sim successfully reproduces global qualitative deformation regimes and dominant bending modes. While this level of fidelity is promising for qualitative task planning, dedicated contact and porosity modeling remain necessary for applications requiring high-precision quantitative prediction.**

***Index Terms*—Robotic manipulation, Deformable objects, Physics simulation, Isaac Sim, Sim-to-real validation**

I. Introduction

Robotic manipulation has become an essential component of modern industrial automation, enabling increased productivity, repeatability, and operational safety across a wide range of manufacturing and logistics processes [1]. While the manipulation of rigid objects has reached a high level of maturity, handling deformable materials remains a challenging problem due to their complex mechanical behavior, high-dimensional state space, and sensitivity to contact conditions [2]. Among deformable objects, paper sheets are particularly relevant because they are widely used in applications such as document processing and office automation, where reliable pick-and-place operations are required [3], [4].

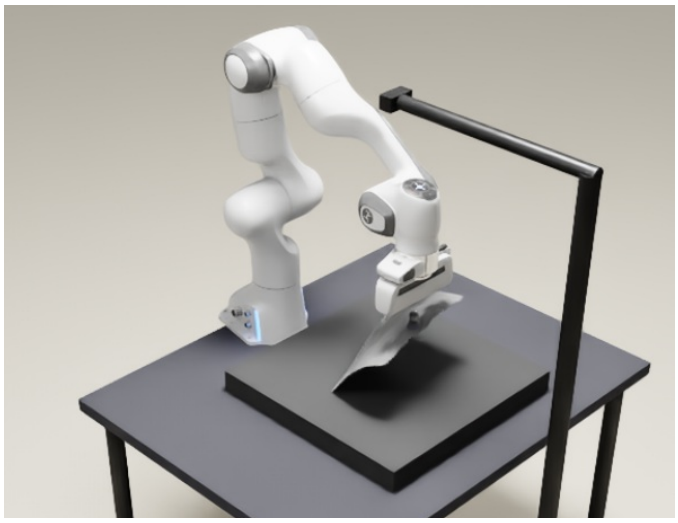


Figure 1. Workbench scene simulated in Isaac Sim: Franka Panda with the pneumatic suction cup on the end-effector, A4 paper sheet on the elevated platform, and zenithal RGB-D camera above.

Manipulating paper sheets presents several technical challenges. Their low stiffness, small thickness, and high flexibility make them susceptible to bending, folding, and slipping during grasping, distinguishing paper from cloth and fabric, for which state-of-the-art manipulation methods have been shown to perform poorly when transferred to paper [5]. A related and distinct challenge is unintended adhesion between multiple sheets during grasping, addressed through friction-uncertainty-based separation strategies [3]. Vacuum suction grippers have become one of the most widely adopted solutions for handling paper because they provide a simple and effective grasping mechanism for lightweight flat objects [4]. However, despite their extensive industrial use, accurately modeling the interaction between suction cups and deformable materials remains difficult due to the complex contact mechanics involved.

Recent advances in physics-based simulation have enabled the development of digital twins for robotic systems, allowing researchers to design, validate, and optimize robotic applications in virtual environments before deploying them on

physical platforms [6]. These approaches reduce development costs, accelerate prototyping, and improve system reliability by minimizing the need for extensive real-world experimentation. Among the available simulation platforms, NVIDIA Isaac Sim has emerged as a powerful framework for robotics research by combining physically based simulation, high-fidelity rendering, and seamless integration with modern robotics software stacks.

Despite these advances, accurately simulating the manipulation of deformable objects remains an open challenge [6]. The behavior of paper during grasping depends not only on its material properties but also on the interaction between the suction cup, the generated vacuum force, friction, and contact dynamics. Consequently, assessing the fidelity of simulation environments requires systematic comparisons between simulated and real robotic systems under comparable operating conditions.

In this work, we investigate the manipulation of paper sheets using vacuum suction in both real and simulated environments. A physical experimental setup was developed using a vacuum gripper, while a digital twin was implemented in NVIDIA Isaac Sim by reproducing the workspace geometry, and physical interaction parameters. Manipulation experiments with the same grasp locations and nominal lift height were performed in both environments, enabling a quantitative comparison of the final static paper deformation.

The main contributions of this work are threefold: (1) the development of a digital twin for vacuum-based paper manipulation in NVIDIA Isaac Sim; (2) the implementation of a corresponding real-world experimental platform for real-to-simulation comparison; and (3) a first real-to-simulation evaluation, including quantitative and qualitative analyses, that assesses the fidelity of paper deformation and identifies key limitations of current simulation models for vacuum manipulation of deformable paper objects.

II. Related Work

a) Physics-Based Simulation for Robotics: Physics-based robot simulation reached a new level with the release of the Newton engine: an open-source, GPU-accelerated engine built on NVIDIA Warp and integrated with OpenUSD, developed through a collaborative effort by NVIDIA, Google DeepMind, and Disney Research. The engine integrates MuJoCo Warp as its primary backend and adopts a modular, solver-agnostic architecture that accommodates diverse materials, including a Vertex Block Descent (VBD) solver for cloth [6]. This new engine generation, Isaac Sim 6.0/Newton, enables the present work. Despite this advance, adopting Newton engine for thin deformable objects still exposes known stability issues. Recent work reports that Isaac Sim with PBD-based cloth is unstable when holding a garment, while Newton VBD produces cloth-rigid body penetration during bimanual manipulation [7]. IsaacIPC addresses this same limitation through a higher-fidelity contact solver, coupling Isaac Sim/Isaac Lab's parallel-environment infrastructure to GPU-accelerated contact simulation from libuipc [8]. Thus, there is a growing demand for studies that quantify the reality gap in sim-to-real applications by systematically comparing simulated and real-world experiments, thereby establishing the validity of the simulation [9], [10].

b) Robotic Manipulation of Deformable Objects: Deformable object manipulation is recognized as a main bottleneck for robotic autonomy in unstructured environments. The infinite-dimensional state space, severe self-occlusions, and complex dynamics of these objects impose significant barriers to perception, modeling, and control [2]. Within this bottleneck, thin paper-like sheets form a subgroup receiving comparatively little attention, generally overshadowed by the cloth/rope pair. Paper is mechanically distinct from cloth and fabric: it possesses significantly higher bending stiffness and a slippery surface, and state-of-the-art cloth/fabric manipulation methods have been shown to perform poorly when transferred to paper [5]. This difficulty is compounded by a purely geometric constraint: the near-zero thickness of a sheet makes direct top-down grasping intrinsically unreliable, forcing manipulation strategies to first induce a curvature or fold via frictional contact before any stable grasp can be formed [11]. Singulation and grasping of individual sheets from a stack have also been investigated using non-suction end-effectors [12], [13]. In contrast, suction-based approaches for paper singulation and grasping remain comparatively underexplored. While vacuum-based manipulation has been explored for other deformable materials (as discussed in the following paragraph), systematic studies on suction-based paper manipulation remain limited.

c) Vacuum-Based Paper Manipulation: Vacuum-based manipulation of thin deformable materials remains challenging due to the complexity of modeling suction contact. Existing work has proposed physically motivated suction models that capture cup deformation and seal formation for real-time simulation [14], while other approaches focus on robust suction grasp planning using analytic models and deep learning [15]. Modern simulation frameworks, such as Isaac Sim, further model suction through rigid constraints and compliant bellows representations [16]. Despite these advances, suction models primarily address grasp mechanics rather than the deformation of thin objects after grasping. Recent studies have also explored vacuum-based manipulation of deformable materials, including textile handling [17] and pressure-controlled singulation of porous sheets [18]. However, these works focus on physical systems, and their modeling principles have not been systematically incorporated into general-purpose simulators. Consequently, there is still no quantitative validation of how accurately simulation reproduces the deformation of thin paper during vacuum-based manipulation. This work addresses this gap through a position-by-position geometric comparison between a digital twin and real-world experiments.

III. Methodology

Our methodology quantifies the sim-to-real gap in vacuum-based paper manipulation through systematic comparisons between simulated and real-world experiments. Specifically, we

evaluate the geometric fidelity of the simulation by comparing, position by position, the 3D deformation of a single A4 paper sheet lifted by a vacuum suction cup in both environments. As illustrated in Fig. 2, the proposed pipeline comprises four stages: *Simulation Capture*, *Real Capture*, *Grasping*, and *Real vs. Simulated Deformation Comparison*. The following subsections describe each stage in detail.

A. Simulation Capture

The simulated environment is implemented in **Isaac Sim 6.0** using the Newton physics engine. As shown in Fig. 1, the scene comprises five main components.

The manipulator is a **Franka Emika Panda** 7-DOF robotic arm. The Panda is used as a kinematic carrier of the suction cup, reflecting the intended robotic integration. The end-effector is a **single-bellow pneumatic suction cup** (Schmalz FMP-30 or equivalent, $\sim$30 mm diameter), comparable to the SFB1 30 cup used in the real setup (Section III-B), modeled as a kinematic constraint on the mesh nodes within the cup footprint (no pressure or airflow is simulated).

Perception is provided by a **zenithal RGB-D camera** (Intel RealSense D415, 1280×720), positioned approximately 60 cm above the paper with its optical axis aligned vertically. The same configuration is reproduced in the real setup, ensuring consistent viewpoints for point cloud comparison.

The workbench consists of a flat 1.0×0.8 m surface supporting a $50 \times 50 \times 5$ cm elevated platform, which positions the paper at approximately 0.80 m above the ground. This configuration facilitates paper segmentation and provides sufficient clearance during lifting. The top surface of this platform is the support plane used as height reference (Section III-D); in the real setup it corresponds to the top of the foam base (Section III-B).

The manipulated object is a single A4 paper sheet (210×297 mm, 75 g/m^2), modeled as a **surface finite element (FEM) shell**. This representation was chosen because paper exhibits high in-plane stiffness but relatively low bending stiffness. Unlike volumetric FEM, which couples stretching and bending through a single Young's modulus, the Newton shell model independently controls these behaviors via separate stretch and bend stiffness parameters, enabling more realistic deformation. The paper is instantiated using `DeformablePrim` with a `SurfaceDeformableMaterial`.

B. Real Capture

The physical setup replicates the simulated workbench to enable a direct sim-to-real comparison. An **Intel RealSense D415** RGB-D camera was mounted approximately 60 cm above the paper with its optical axis aligned vertically, matching the simulated camera configuration. A single A4 paper sheet is grasped and lifted using a SCHMALZ SFB1 30 SI-55 SC040 pneumatic suction cup.

The system was mounted on a flat wooden table. A 50x50x5 cm D33 density foam base was used to prevent suction issues with the smooth tabletop, where the sheet of paper is positioned centered under the D415 camera, which is mounted 60 cm high relative to this base on an ELGATO FLEXARM mount. The suction cup was attached to the shaft of a long screwdriver and connected to a ZR370-02PM mini vacuum pump powered by four 1.5V AA batteries and an on/off switch.

For each grasp, the system records the RGB image, depth map, point cloud, and camera intrinsic parameters provided by the RealSense SDK. The D415 is factory calibrated, and the depth image is aligned with the RGB image, allowing direct back-projection of image pixels into 3D space using the camera intrinsics without additional calibration or image rectification.

C. Grasping

Real grasp mechanics: Upon touching the sheet of paper with the suction cup, the on/off switch is activated and the sheet is grasped via suction at the selected contact point, being lifted to approximately 25 cm (visually confirmed with a ruler during the lifting process). Once the lifting is confirmed, the spacebar is pressed, triggering the image capture process by the D415 camera, followed by turning off the on/off switch.

Simulated grasp mechanics: After contact, the mesh nodes within the suction cup footprint are pinned to the end-effector by updating their target positions at every physics step using `set_nodal_positions`. As the suction cup lifts, the pinned nodes follow its motion while the remaining nodes deform freely under gravity, producing the paper bending. To ensure stable attachment, a smoothstep interpolation is applied during pinning and the velocity of the pinned nodes is reset to prevent numerical instabilities. The simulation is executed using Isaac Sim's GPU pipeline, where physics updates are not propagated back to the USD stage. Consequently, robot joints and deformable nodes are controlled through direct API calls (`set_joint_positions` and `set_nodal_positions`), while the end-effector pose is computed via forward kinematics using the Lula library. Following this procedure, the robot approaches the selected grasp point, establishes the suction attachment, and lifts the paper approximately 25 cm.

D. Real vs. Simulated Deformation Comparison

The proposed comparison strategy evaluates the geometric fidelity of the simulation by comparing the paper deformation produced at a set of predefined grasp locations. Nine grasp points are considered: the sheet center, the four corners, and the midpoints of the four edges. This configuration samples the principal deformation regimes while ensuring that nominally the same geometric contact locations are used in both the real and simulated environments.

Before comparison, the paper is isolated from the scene using the same processing pipeline for both real and simulated point clouds. Implemented with Open3D, NumPy, and OpenCV, the pipeline consists of (i) voxel downsampling (2 mm), (ii) HSV color filtering to retain the paper while removing the grasping tool, and (iii) extraction of the largest connected component using DBSCAN [19].

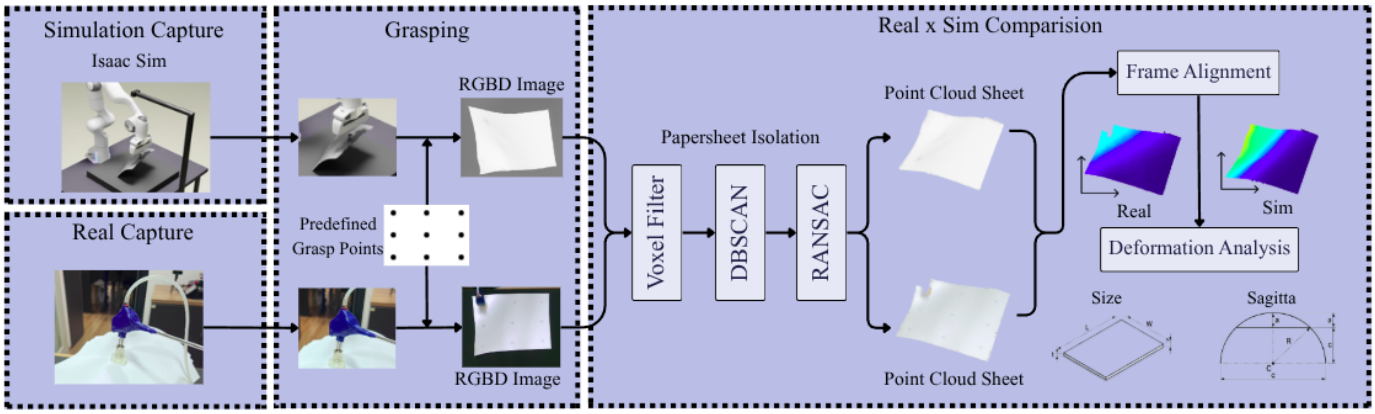


Figure 2. Overview of the methodology pipeline: the simulated and real captures produce point clouds that are isolated to the paper sheet, obtained under a common vacuum-suction grasp, and finally compared in a shared reference frame.

The isolated point clouds are then transformed into a common reference frame. The support plane is estimated by deterministic RANSAC [20] followed by least-squares refinement. The plane defines the origin and the vertical ($+Z$) axis, while the in-plane X and Y axes are aligned with the paper through Principal Component Analysis (PCA). This normalization removes differences in camera pose and coordinate conventions between the real and simulated acquisitions.

Finally, the paper deformation is quantified using three geometric metrics: the **sagitta**, computed as the difference between the 98th and 2nd percentiles of the height distribution as a robust height-range descriptor of the deformed sheet; the **planar dimensions**, measured along the PCA-aligned principal axes; and the **maximum height** of the sheet above the support plane. These metrics are computed identically for both datasets; the sagitta and the maximum height are used to quantify the sim-to-real agreement, whereas the planar dimensions are reported as a coarse descriptor only (Section IV).

Note that the sagitta, defined here as the 98th–2nd percentile range of heights, is a height-range descriptor rather than a geometric sagitta; when the lowest part of the sheet touches the support plane, it approaches the maximum height, so the two metrics are not independent. Moreover, PCA-aligned extents depend on the sheet's footprint and may not coincide with its edges.

IV. Experiments and Results

A. Paper Simulation Experiments

The simulated paper model was tuned by evaluating multiple combinations of material representation, stiffness parameters, mesh resolution, grasp implementation, and solver iterations in the Newton engine. Table I summarizes the tested configurations and their qualitative behavior. The adopted configuration corresponds to Row 7, which provided the closest visual agreement with the real paper deformation at the Center grasp. A key outcome of this tuning process is that increasing the number of solver iterations from 32 to 250 had a larger impact on reproducing paper-like behavior than increasing the bend stiffness alone, enabling a smooth single-curvature deformation instead of the fine wrinkles characteristic of cloth-like materials.

Table I
Parameter configurations evaluated for the simulated paper in Isaac Sim (Newton engine). The qualitative behavior of each configuration is reported. Row 7 corresponds to the configuration adopted in this work.

#	Model	E (Pa)	Stretch / Bend	Iter.	Observed Behavior
1	Volumetric	10^9	—	100	Excessive stretching
2	Volumetric	10^{12}	—	100	Excessive rigidity
3	Surface	10^{11}	10^4 / 10^3	32	Cloth-like deformation
4	Surface[a]	10^{11}	10^4 / 10^3	250	Cloth-like deformation with wrinkles
5	Surface	10^{11}	10^4 / 1.5×10^4	32	Insufficient bending convergence
6	Surface[b]	10^{11}	10^4 / 1.5×10^4	250	Overly rigid; nearly planar lifting
7	**Surface[c]**	10^{11}	10^4 / 1.5×10^4	**250**	**Closest match to real paper**

[a] Fine mesh (60×42); all other configurations use a 24×17 mesh.
[b] Grasp implemented using the built-in deformable attachment (`create_auto_deformable_attachment`).
[c] Grasp implemented by direct node pinning (`set_nodal_positions`); adopted configuration.

B. Real-to-Sim Deformation Comparison

The nine grasp positions were compared in the common reference frame defined in Section III-D. We evaluate the agreement between the real and simulated sheets using both quantitative metrics and qualitative deformation patterns.

1) Quantitative results: Table II summarizes, for each position, the three robust metrics of Section III-D — planar dimension, sagitta f, and top height $z_{\max}$ — for the real and simulated sheet, together with the simulation-to-real deviations Δf and Δz (planar dimensions are reported as a coarse descriptor only). Each real value comes from a single trial, and the Center grasp (†) is the one used for calibration.

Planar dimension. The nominal dimensions of the A4 sheet are 297×210 mm, but several planar bounding dimensions

Table II

REAL VS. SIMULATED METRICS ACROSS THE NINE GRASP POSITIONS (ALL VALUES IN MM). f: SAGITTA (98TH–2ND PERCENTILE HEIGHT RANGE); $z_{\max}$: MAXIMUM HEIGHT ABOVE THE SUPPORT PLANE; $\Delta = \text{sim} - \text{real}$. †GRASP USED FOR CALIBRATION; THE REMAINING EIGHT POSITIONS WERE HELD OUT. REAL VALUES ARE SINGLE TRIALS.

	Planar Dimension		Sagitta f			Maximum Height		
Position	**Real**	**Sim**	**Real**	**Sim**	Δf	**Real**	**Sim**	Δz
Center†	307×188	299×246	64	99	$+35$	147	132	-15
Upper-left c.	294×259	343×277	171	150	-21	179	158	-22
Upper-right c.	345×254	311×246	136	147	$+11$	139	160	$+20$
Lower-left c.	349×243	321×258	138	143	$+6$	148	156	$+7$
Lower-right c.	341×245	341×275	106	156	$+50$	113	168	$+55$
Upper edge	317×168	312×175	128	141	$+13$	142	165	$+23$
Lower edge	304×176	307×171	107	137	$+30$	124	162	$+38$
Left edge	306×275	306×238	160	119	-41	169	137	-32
Right edge	320×297	280×234	157	133	-24	166	138	-28
MAE (9 positions)	—		—		**26**	—		**27**
MAE (8 held-out)	—		—		**24.5**	—		**28.1**

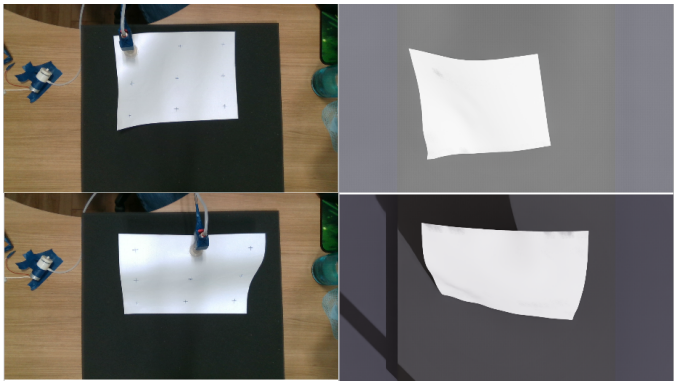


Figure 3. Qualitative comparison between real and simulated paper deformation for representative cases. The first row shows the corner grasp configurations, while the second row shows the edge grasp configurations. In both rows, the real experiment is shown in the left column and the corresponding simulation is shown in the right column.

estimated from the point clouds exceed these values in both the real and the simulated setups (e.g., 349×243 mm real and 343×277 mm simulated; the sheet diagonal is ≈ 364 mm). As expected, the extent along the primary bending axis decreases with the curvature severity (e.g., Y-extent of ≈ 168–176 mm for upper/lower edge grasps). Because the extents are measured along PCA-aligned axes of a curved, lifted sheet, they need not coincide with the sheet edges and may be inflated, particularly for nearly isotropic footprints; depth-edge artifacts and residual outliers in the real point clouds may also contribute. We did not isolate the cause of this overestimation. Therefore, planar dimensions are reported for completeness and are not used as a fidelity metric; our conclusions rely on sagitta and maximum height (Section IV-D).

Overall metric agreement. The mean absolute error (MAE) between simulation and reality is 26 mm for the sagitta and 27 mm for the maximum height (24.5 and 28.1 mm, respectively, over the eight held-out positions). Considering the observed deformation amplitudes (106–171 mm), these errors correspond to relative deviations of approximately 15–20%.

For the sagitta, three grasp positions produce lower simulated values, while six produce higher values; the errors therefore do not share a single sign, although a systematic component cannot be excluded.

Best and worst cases. The closest agreement is observed for the lower-left corner, upper-right corner, and upper edge grasps, with sagitta deviations of only $+6$ mm, $+11$ mm, and $+13$ mm, respectively. In contrast, the largest discrepancies occur for the lower-right corner ($\Delta f = +50$ mm) and left edge ($\Delta f = -41$ mm). Since these errors occur in opposite directions, they suggest variability in the grasp configuration rather than a systematic limitation of the simulator.

2) Qualitative Evaluation: Beyond the numerical measurements, the simulated sheet reproduces the main deformation regimes observed in the real experiments. Figure 3 presents representative comparisons for corner and edge grasp configurations. The qualitative agreement suggests that the simulator reproduces the characteristic deformation behavior of the paper, with similar bending directions, curvature profiles, and overall shapes across the evaluated grasp configurations.

C. Additional Observations and Simulation Limitations

The real-world experiments revealed two relevant phenomena that are not accurately reproduced by the current simulation setup. These observations highlight physical interactions that are important for paper manipulation but are typically neglected in rigid-body and deformable-body simulators.

i) Vacuum interaction with porous paper surfaces. Paper is a porous material that behaves similarly to a micro-perforated membrane. Consequently, lifting a single sheet from a paper stack using a suction cup is a non-trivial process, since the vacuum pressure must be carefully adjusted to avoid simultaneously attaching multiple sheets. This effect is strongly related to the air permeability of paper and does not occur with impermeable materials such as plastic sheets.

In the Isaac Sim environment, this behavior is not realistically reproduced. A single simulated sheet can be lifted from a stack without difficulty, since the interaction between the suction pressure and the porous structure of paper is not modeled. Furthermore, in our implementation, the suction cup is constrained to the sheet surface by directly binding mesh points, which simplifies the grasping interaction but prevents the simulation from capturing the gradual pressure-driven attachment observed in the real setup (Fig. 4). A more accurate simulation would require modeling airflow, permeability, and the interaction between adjacent sheets.

ii) Influence of the supporting surface. The physical properties of the supporting table also significantly affect the manipulation process. In the real experiments, the sheet must be placed on a rough or slightly compliant surface, such as a firm sponge, to prevent undesired suction between the cup and the table. When the paper stack is placed on a smooth and impermeable surface, the suction cup can seal against the table during contact, causing the entire stack or support surface to be lifted. This phenomenon is not observed in the simulated environment, where the contact interaction between the suction cup and the supporting surface does not reproduce

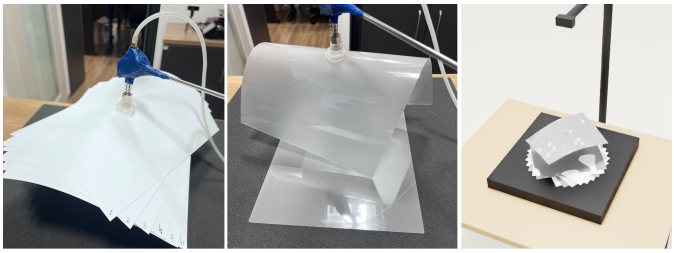


Figure 4. Real-world observations of paper manipulation: multiple sheets lifted by a suction cup, different behavior of plastic sheets, and the corresponding non-realistic behavior observed in Isaac Sim for stacked paper sheets.

the sealing effects caused by real vacuum forces. Therefore, accurate simulation of paper manipulation requires not only modeling sheet deformation, but also the interaction between vacuum, material permeability, and the contact properties of the supporting surface.

D. Limitations

This study provides a preliminary qualitative assessment of the simulated paper deformation. Differences between the controlled simulated actuation and the manually performed real lifting, together with the lack of repeated trials, limit the separation of simulation error from experimental variability.

V. Conclusion

This work presented a real-to-simulation comparison of paper manipulation using a vacuum suction gripper. A physical experimental platform and a corresponding digital twin were developed in NVIDIA Isaac Sim, enabling the evaluation of equivalent manipulation tasks under controlled conditions.

The results indicate that, after parameter tuning, the simulator reproduces the overall deformation behavior of paper, with mean absolute errors of 26mm in sagitta and 27mm in maximum height across nine grasp configurations. It also captures the characteristic deformation modes of corner, edge, and center grasps. However, the current framework does not accurately model vacuum–paper interactions or the supporting surface, limiting its realism, particularly when manipulating paper stacks.

Overall, this study indicates that physics-based simulation can serve as a useful tool for qualitative studies of robotic paper manipulation, while also identifying the physical phenomena that require more accurate modeling. Future work includes repeated and instrumented real trials to report variability, quantitative calibration with held-out validation, full point-cloud metrics and spatial error maps, incorporating airflow and material permeability models, improving suction-contact simulation, evaluating multiple paper types and thicknesses, and extending the proposed digital twin to more complex manipulation tasks such as page turning, sheet separation, and document handling.

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